PORTFOLIO 5: DISTRIBUTED AND CLOUD DATABASE EXERCISE

Case Study: Library Management System.
Database: SQL Server.
Proposed Cloud Platform: Microsoft Azure SQL Database.

1. Introduction

The purpose of this project is to explore how distributed and cloud databases can improve the operation of a Library Management System.

A distributed database allows information to be stored and accessed across multiple locations. This is useful for a library with branches in different cities because each branch needs reliable access to books, members, and borrowing records.

2. Proposed Distributed Database Architecture

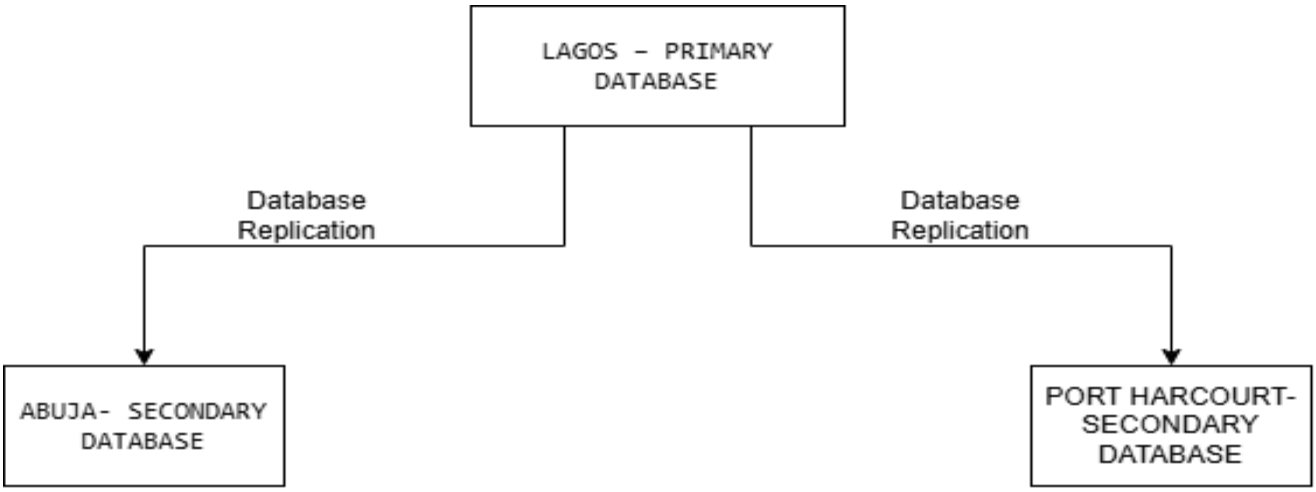
The proposed system consists of three library branches:

- Lagos – Primary database.
- Abuja – Secondary database.
- Port Harcourt – Secondary database.

The Lagos database serves as the primary location, while the other branches receive replicated information.

This arrangement allows the branches to access library information without depending entirely on one physical location.

Figure 1: Distributed Database Architecture



SQL Server- Database Management System

3. Database Replication

Database replication involves copying information from one database server to another.

In the proposed system, Lagos acts as the Publisher, while Abuja and Port Harcourt act as Subscribers.

When new book or member information is added in Lagos, the changes can be replicated to the other branches.

This helps keep information available across different locations.

4. Database Sharding

Sharding involves dividing a large database into smaller sections and distributing them across multiple servers.

For the Library Management System, member records could be distributed according to their registered branches.

For example:

Database Location	Records
Lagos	Lagos members
Abuja	Abuja members

Port Harcourt Port Harcourt members

This can improve data management and reduce the workload on a single database server.

5. Availability and Consistency

Availability: This means ensuring that users can access the database when needed. If the Lagos server becomes unavailable, the other branches may continue accessing their local replicated data, depending on the system configuration.

Consistency: This ensures that information remains accurate across the different database locations. For example, when a book is borrowed, the system should ensure that its availability is updated correctly to prevent conflicting borrowing records.

6. Cloud Database Deployment

Cloud database deployment involves hosting database services on cloud infrastructure instead of relying entirely on physical servers.

Microsoft Azure SQL Database is a suitable example for this project.

The library could host its database in Azure, allowing authorised users from different branches to access information through an internet connection.

Cloud deployment also provides options for automated backups, scalability and database monitoring.

7. Benefits of the Proposed System

The proposed distributed and cloud database system offers several benefits:

- Improved access to library information across branches.
- Reduced dependence on a single physical database server.
- Better data availability.
- Easier expansion as the library grows.
- Centralised management of database services through cloud infrastructure.

8. Conclusion

This project demonstrates how distributed database concepts can be applied to a Library Management System.

Replication, sharding, availability and consistency can help improve database management across multiple branches. Cloud deployment using Microsoft Azure SQL Database also provides opportunities for scalability and easier access.